

# Empirical Study of Memorization in Differentially Privately Fine-Tuned Large Language Models

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[Link to GitHub Repo](#)

## Abstract

### 1. Introduction

In 2017, a group of researchers at Google Brain proudly announced the architecture of the transformer neural model, which revolutionized the field of Natural Language Processing (NLP) and led to the rise of large language models (LLMs) (Vaswani et al., 2017). Training LLMs requires large amounts of data, and if not curated carefully, these datasets might contain sensitive information such as social security numbers or medical records. This poses a severe problem in model security and ethics, as shown by Carlini et al. (2019) that neural networks have a high risk of unintentionally memorizing fine-grained information and oddly-specific details. To mitigate this issue and protect the privacy of individuals, one approach is to leverage methods from *differential privacy* (DP), which provides a rigorous mathematical framework for quantifying privacy (Dwork et al., 2006). With DP, researchers have developed DP-based training methods, such as DP-SGD, to train neural networks and showed that they effectively mitigate privacy concerns during training. Background information about DP and DP-SGD utilized in this paper is described in Section 2.

Despite the prominent performance of DP-based training methods, the use of these methods in training or fine-tuning LLMs still remains underexplored in both research and industrial settings. Specifically, the benefits of DP in reducing unintentional memorization of training data are to be precisely quantified. Therefore, this study aims to narrow down this gap by investigating the following research question:

*At what privacy budget  $\epsilon$  does DP-SGD effectively prevent memorization of sensitive sequences in fine-tuned LLMs, and what is the accuracy cost of that protection?*

### 2. Background

Differential privacy is a mathematical framework that leverages probability theories to rigorously quantify privacy in algorithms (Dwork et al., 2006). Formally, let  $\mathcal{X}$  denote the data space and  $X = (x_1, \dots, x_n) \in \mathcal{X}^n$  be a dataset. We can then define another dataset  $X' = (x'_1, \dots, x'_n) \in \mathcal{X}^n$  to be a *neighboring dataset* of  $X$ ,

denoted as  $X \sim X'$ , if there exists  $i \in \{1, \dots, n\}$  such that  $x_i \neq x'_i$ , and for all  $j \neq i$ ,  $x_j = x'_j$ . With this setup, we say that a randomized algorithm  $\mathcal{M}$  is  $(\epsilon, \delta)$ -*differentially private* if for all datasets  $X \sim X'$ , and all  $S \subseteq \Omega$  where  $\Omega$  is the output space of  $\mathcal{M}$ ,

$$\mathbb{P}(\mathcal{M}(X) \in S) \leq e^\epsilon \mathbb{P}(\mathcal{M}(X') \in S) + \delta.$$

In the context of training deep neural networks, we can adapt this definition and make gradient descent a differentially private algorithm so that the trained models do not memorize specific details of any training data. Defining  $\mathcal{L}$  as the loss function,  $\alpha$  as the learning rate, instead of updating the weights by  $\alpha \nabla \mathcal{L}$  at every step, we first compute each per-sample gradient  $\nabla \mathcal{L}_i$  and clip it to the L2 norm bound  $C$ , then sum the clipped per-sample gradients and add random noise from  $\mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$  to the sum before updating. By choosing the parameters  $\sigma$  and  $C$  strategically, one can show that this algorithm, named *differentially private stochastic gradient descent* (DP-SGD), satisfies  $(\epsilon, \delta)$ -differential privacy (Abadi et al., 2016).

### 3. Related Work

Since the introduction of the concept of DP, most previous work focused on using DP techniques on training or fine-tuning neural networks of simpler architectures compared to transformers. For example, on language modeling, McMahan et al. (2018) applied a DP algorithm called *federated averaging* (FedAvg) on training recurrent language models such as RNNs and LSTMs. In the study, they experimented with various hyperparameters for noise and gradient clipping to quantify the downside of DP training on the performance of language models. On the other hand, the study by Carlini et al. (2019) conducted experiments on investigating language modeling ability of LSTM models on the Penn Treebank dataset. A similar characteristic shared across these two papers is that the models they used in their experiments are rather small: 600,000 parameters for the model in Carlini et al. (2019) and 1.35M parameters for the model in McMahan et al. (2018). Since LLMs have grown exponentially in size nowadays, there is an urgent need for more research on deploying DP techniques on training larger models.

Studies on applying DP techniques on fine-tuning LLMs do exist, with Yu et al. (2024) being a foundational one. In this paper, the authors conducted two experiments, one on natural language understanding (NLU) and the other on natural language generation (NLG), both using DP-SGD, and they compared the model performance against the state-of-the-art on different benchmarks. The models they used come from the GPT-2 and RoBERTa suites, with GPT-2 Small being the smallest model but still having 117M parameters which is much larger than the models in previous studies. More recently, a paper by Du et al. (2025) also examines the effect of DP-SGD in fine-tuning GPT-2 models. However, they focused more on comparing different fine-tuning methods, including FFT, LoRA, prefix-tuning, and P-tuning, rather than directly investigating DP-SGD on GPT-2.

Despite these studies on DP fine-tuning LLMs, the datasets they used to evaluate their models are not inherently private: Yu et al. (2024) used data from the restaurant domain and DART which is open-domain,

while Du et al. (2025) used Wikitext-2-v1 and AG News. The downside is that all these datasets are public sources and do not contain sensitive information. In order to understand how DP techniques perform on private datasets, which was the reason DP was developed in the first place, it is preferred to apply them to a more private, sensitive domain so that we can truly evaluate the effectiveness of DP before its actual deployment in real-world settings. This is thus the main motivation of this project: applying DP techniques to a mental health dataset, which is inherently privacy-sensitive.

## 4. Methodology

### 4.1. Model

In this project, we used the GPT-2 Small model developed by OpenAI (Radford et al., 2019). GPT-2 Small is a decoder transformer model that transforms an input directly to the output, unlike BERT which outputs embeddings of the input sequence. As such, GPT-2 is known for its ability in language generation, which matches the objective of this study. The GPT-2 Small model consists of an embedding layer with positional encoding, 12 transformer blocks, and an output layer, adding up to 117M parameters and thus is a great balance of performance and computational cost.

### 4.2. Data

The data used in this project is a [Reddit Mental Health Dataset](#) available for downloads on Hugging Face. There are 151,228 rows in total in the dataset, each row being a post related to mental health containing attributes such as the user, the text of the post, the time posted, and more. Only the text of the posts is of interest in this project. Other attributes are discarded. From the raw dataset, we randomly sampled 10,000 posts to form the fine-tuning set, and 1,000 more as the validation set.

### 4.3. Experimental Design

In the study by Carlini et al. (2019), the authors proposed the method of *canary insertion*. In this context, a *canary* refers to a piece of text purposefully inserted into the training data in order to measure the extent to which the model memorizes the training data instead of learning the underlying patterns. In their study, they used canaries of the form

*The random number is ...*

and fill in the blank with 9-digit numbers. By varying the number of times the canaries were inserted and the privacy budget  $\epsilon$  in DP-SGD, the authors were able to quantify how much DP helps with reducing unintentional memorization of the training data.

In the current project, we adopt the same design and apply it to our mental health dataset with the following form of canaries:

*My patient ID is ...*

and fill in the blank with a random 6-digit number representing a patient ID. The reason for choosing this canary form is that it is directly related to mental health, which is the theme of the dataset, and is inherently private information that a patient would not wish to reveal which is the main motivation of the project.

With this canary form, we vary the number of times the canaries are inserted into the training set, with  $\text{freq} = 1, 5, 10, 50$ . We also pre-specified some values of privacy budget to conduct the experiment, with  $\epsilon = 0.5, 1, 2, 4, 8, \infty$ , where  $\epsilon = \infty$  means vanilla gradient descent. For each pair of frequency and privacy budget  $\epsilon$ , we fine-tuned GPT-2 Small using the training set with canaries inserted for 3 epochs. The Python library `Opacus` was used for DP-SGD support. Each model was then assessed by calculating the *log-perplexity* on the validation set as a proxy for model utility, mathematically expressed as

$$\text{log-perp}_{\theta}(x_1, \dots, x_n) = -\frac{1}{n} \sum_{i=1}^n \log_2 \mathbb{P}(x_i | f_{\theta}(x_1, \dots, x_{i-1})).$$

To see more fine-grained differences between different privacy budget  $\epsilon$ , we report the validation *perplexity* instead, expressed as

$$\text{perp}_{\theta}(x_1, \dots, x_n) = \exp(\text{log-perp}_{\theta}(x_1, \dots, x_n)).$$

To measure unintentional memorization of the training data, we also computed the *exposure* of each model as

$$\text{exposure} = \log_2(900000) - \log_2(\text{rank}),$$

where  $\text{rank}$  measures the rank of the probability that the random number in the canaries appears among all 900,000 possible 6-digit numbers. A small exposure means that the model does not memorize the canaries and the output distribution is generated entirely by chance, while  $\text{rank} = 1$  and  $\text{exposure} = \log_2(900000) \approx 19.8$  indicates that the model memorizes the canaries perfectly (Carlini et al., 2019).

It is worth noting that not all parameters in the GPT-2 Small model were fine-tuned. In the project, we froze the embedding & positional encoding layer as well as the transformer head, and only fine-tuned the 12 transformer blocks. This was necessary due to GPU memory constraints, consistent with the approach of Yu et al. (2024).

## 5. Results & Discussion

## 6. Conclusion

### 6.1. Limitations & Future Directions

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